

Project Design Phase-II Technology Stack (Architecture & Stack)

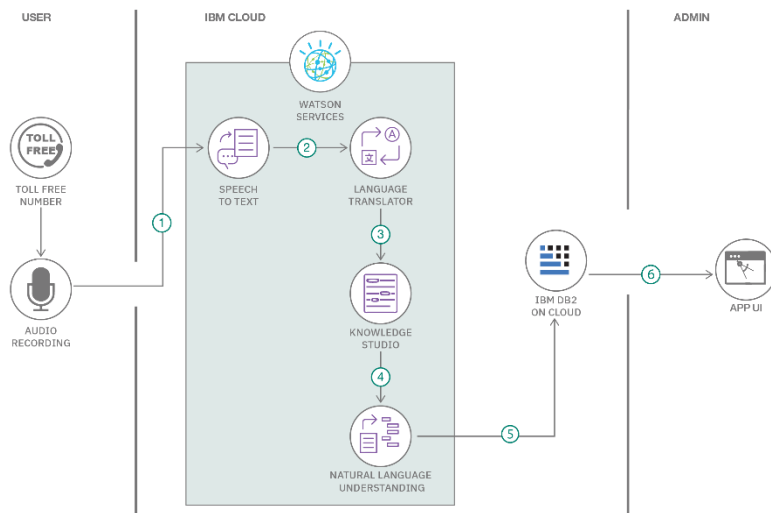
Date	31 January 3035
Team ID	LTVIP2025TMID32230
Project Name	SmartSDLC – AI-Enhanced Software Development Lifecycle
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
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1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used

S.No	Characteristics	Description	Technology
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

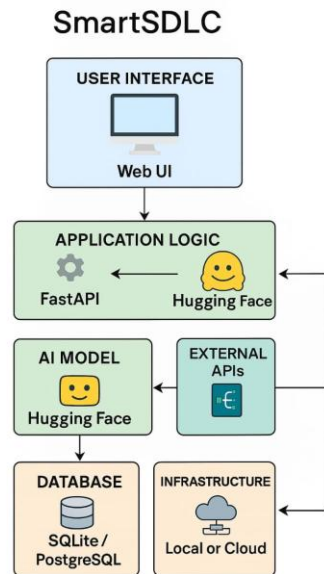
<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

SmartSDLC – AI-Enhanced Software Development Lifecycle

📄 Technical Architecture Diagram



Frontend/UI Layer: User interacts through a web app (Streamlit).

Backend Logic: FastAPI handles the request-response lifecycle.

AI Models (LLMs): Hugging Face APIs process natural language for requirements extraction, code generation, and test generation.

Database: Stores uploaded documents, processed outputs, user data.

External APIs: Integration with GitHub, cloud storage, or authentication.

Infrastructure: Deployed on local systems or cloud platforms (AWS, Azure, or IBM Cloud).

 Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web UI where users upload documents and receive outputs	HTML, CSS, Streamlit
2	Application Logic-1	Upload and process document	Python + FastAPI
3	Application Logic-2	Extract requirements using NLP models	Hugging Face Transformers
4	Application Logic-3	Generate code/test cases using AI	Hugging Face GPT-style LLM
5	Database	Store user data, processed content	SQLite / PostgreSQL
6	Cloud Database	Cloud-hosted user and file data (optional)	AWS RDS / IBM Cloudant
7	File Storage	Store uploaded documents	Local Filesystem / AWS S3 / IBM Block Storage
8	External API-1	Model API for code generation	Hugging Face Inference API
9	External API-2	Authentication APIs (OAuth, if any)	Google OAuth / GitHub Login (optional)
10	Machine Learning Model	NLP model for understanding SDLC content	BERT / GPT-2 / CodeT5
11	Infrastructure (Server)	Cloud or local deployment	Localhost, AWS EC2, Render, or Streamlit Cloud

 Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Python frameworks and NLP libraries	FastAPI, Streamlit, Transformers (Hugging Face)
2	Security Implementations	Basic auth, input validation, API token control, secure document upload	OAuth 2.0, HTTPS, SHA-256 hash, JWT